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A Web-based Reference Architecture for Mobile Learning: Its Quality Aspects and Evaluation

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***Abstract*—This paper discusses the development process and evaluation of a reference software architecture within the field of mobile learning. The architecture implements the workflow of *preparation – activity – analysis & reflection* of (data collection) activities supported by mobile applications. A number of challenges, like portability, flexibility, or expandability are important to address in mobile learning. To overcome the mentioned challenges in this field, we are proposing a web-based architecture that is supposed to serve as a foundation for the development of software solutions in the field of mobile learning. We discuss scenario descriptions needed for the evaluation and present three implementations in this field that serve as candidate architectures. With the support of those candidates, the reference architecture is proposed. Important system quality attributes are identified and then applied for an evaluation of the architecture using the SAAM method. We conclude by discussing that the proposed architecture does not only fit the field of mobile learning but can also be adapted to a general solution regarding the workflow of: *preparation – activity – analysis & reflection* of mobile (data collection) applications.**

Keywords—Software Architecture Evaluation; Reference Architecture; Technology Enhanced Learning; SAAM; Web-based

I. INTRODUCTION

The importance of integrating information and communication technologies (ICT) in the field of education is gaining more and more attention. Educational institutions recognize the trend of digitally supported education and studies show that introducing the latest developments of ICT into classrooms can improve teaching and learning methods [1,2]. Modern pedagogical approaches promote the mobility of learners and therefore, members of the Technology Enhanced Learning (TEL) community advance supporting new activities with modern mobile devices like smartphones or tablets. Those activities and their technological support are referred to as mobile learning [3]. Furthermore, mobile technologies can facilitate learning outside traditional environments, such as classrooms, to enhance the learning experience [4].

Especially mobile devices like smartphones and tablets can be utilized to support mobile learning activities. Positioning technologies, high definition cameras, and other sensors, as well as ubiquitous Internet access became standard features embedded in these devices. However, despite of the multiple benefits this setup offers, a major challenge for developers and researchers of mobile applications arises due to the broad fragmentation of mobile devices and their operating systems. Nowadays, the availability of mobile devices like smartphones or tablets is vast among students. The often applied concept of using available devices in different settings is referred to as "bring-your-own-device" and is especially interesting for the field of mobile learning [5]. Because of the resulting heterogeneity of devices, it is important to offer cross-platform solutions.

Our efforts are driven by the mentioned challenges. Emerging web technologies provide some of the solutions that can be used to address these challenges [6], especially regarding cross-platform solutions. In spite of several advantages of web technologies for mobile devices, researchers and developers do not yet recognize the potential it can have in the field of mobile learning [7]. Thus, the majority of existing software solutions are based on native mobile applications limited to certain platforms.

Furthermore, the field of mobile learning struggles with various implementations based on different technology stacks and/or architectures. The outcome is a great amount of inflexible and incompatible solutions that often do not match the dynamic requirements of this field, therefore missing a common foundation to solve similar challenges. Defining a reference architecture is a well-recognized method to tackle this challenge. According to Wilson et al. [8], a reference architecture is a template solution for a concrete architecture for a certain domain. It is a software architecture where structures, relations, and elements provide solutions for either a particular domain or a group of software systems. Additionally, Martinez et al. [9] mention the fact that reference architectures increase the development speed, reduce operational expenses, and improve quality in software systems. Thus, one goal of this work is to provide the first step towards a reference architecture that aims to tackle the mentioned challenges.

In the remaining of this paper, we will provide an overview of our motivation for carrying out this work. This is followed by a description of different scenarios and the proposal of a reference architecture for a specific problem domain in the field of mobile learning. Thereafter, we will present an evaluation of this architecture following the SAAM method.

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II. Motivation and Research Questions

The continuous evolution of learning technologies combined with the rapid developments within the TEL field result in dynamic, complex requirements that are challenging to meet. As a result, the evolution of web and mobile software, together with the dynamic requirements, and the fact that these systems evolve over time, make it difficult to adapt to the constant evolving conditions [10, 11]. Thus, we aim to provide an architectural foundation in form of a reference architecture that strives to address these challenges.

Web technologies offer a promising approach to tackle many of the mentioned challenges. And particularly regarding mobile devices, they are constantly under progressing development. New standards regarding access of sensors or other mobile specifics like background processes and notifications are continually being published to make mobile web solutions more applicable especially when it comes to cross-platform solutions. As mentioned before, the scenario we are interested in is the "bring-your-own-device" one [5], therefore cross-platform is a desired solution for addressing it. However, while working with web technologies, especially when it comes to mobile development, environments are becoming fragmented; there are multiple browsers, which in different ways comply with web standards; diverse mobile platforms; and physical devices that have varying characteristics [12–14].

The field of TEL faces the challenges to rapidly response and adapt to the needs of users and quick technological changes as the learning ecosystem evolves. This requires the ability to rapidly modify and extend solutions. Therefore, software in this area needs to have the ability to grow and evolve over time. An architecture according to these needs as a foundation is desirable for representing a fundamental organization of a system in terms of the key components and their relationships to each other [12].

New technologies are constantly emerging, existing ones are evolving in a very fast pace, and modern teaching and learning methods are introduced. Thus, a central challenge is to provide a foundation that can deal with this dynamic scenario [15]. To the best of our knowledge there are only few efforts investigating an architectural approach to address these challenges within TEL. Therefore, our aim is to define a reference architecture that serves as a foundation for a problem domain in the TEL field. The focus of this reference architecture is on the usage of mobile devices. Therefore, we are placing this work in the sub-domain of TEL: mobile learning. We conducted a study [16] that shows a great deal of mobile learning activities are strongly related to data collection activities. In mobile learning, one of the most prominent examples for data collection activities is called a *field trip*. Therefore, a field trip serves as the foundation for our architectural design. We based our approach on the method for conducting field trips by Krepel and DuVall [17]. In their work, they define the three phases of: *preparation*, *activity*, *reflection*. We are expanding the method proposed by Krepel and DuVall by adding *analysis* to the reflection phase.

In summary, the interplay between the difficulty to adapt to dynamic requirements and the fragmentation of devices and operating systems in TEL mobile learning environments, leads to the need for systems to have the ability to adapt and grow over time. Due to these fundamental architectural challenges, we are proposing a reference architecture to present a foundation for the development of mobile learning applications. This reference architecture is based on the work of the past three years and our efforts within three different projects dealing with mobile learning challenges. Therefore, our current efforts are directed towards exploring the following research questions:

(RQ1) – What are the key components of a reference architecture to satisfy the constantly evolving requirements in the field of mobile learning?

(RQ2) – What methods are suitable approaches to evaluate a (reference) architecture in the field of mobile learning?

III. Scenario Descriptions and Implementations

In this section, we describe three scenarios and their implementations to discuss possible candidate architectures that serve as the foundation for the evaluation process.

A. mLearn4web

mLearn4web is a system that tackles the challenge of introducing new technologies in educational settings. Some countries like Sweden changed their national curriculum to encourage the inclusion of ICT in educational settings [2]. Thus, in order to take full advantage of the available technological setup – the broad availability of mobile devices – it is necessary to provide support for teachers to compose their own learning applications for those devices. Therefore, we have developed a system that allows users without technical knowledge or developing skills to develop their own mobile applications [18, 19]. This system consists of three components that match with the three components, derived of the theory of a field trip [17]: authoring tool, mobile application, and visualization tool.

The authoring tool allows the design of mobile applications with simple, well-known interaction methods like drag and drop. Teachers can design their mobile applications based on designing screens, similar to Ionic Creator[1]. In the authoring tool, it is possible to create a number of screens, rearrange their order and delete them. Onto those screens, it is possible to add pre-defined content components, like adding text/numerical data, capturing image/video/audio data and recording location data (GPS) through drag and drop. The resulting mobile application is automatically deployed as a mobile web application with the content that was specified before.

The collected data is presented in a visualization tool. In this tool, data that belong to the same screen are considered to be in relation to each other. For instance, if on one screen a picture was taken and on the same screen the location was also stored, it is assumed that those metadata attributes are part of the same scenario object. Therefore, the data is presented in an appropriate way. In this case, a map is presented where

[1]https://creator.ionic.io/

markers indicate that additional data is available, e.g., by clicking on the marker the picture is presented on the map.

B. TriGO

The TriGO project [20] is similar to mLearn4web. However, in contrast to mLearn4web, this project exclusively aims to support learning in the field of mathematics education. The purpose of TriGO is to introduce concepts of trigonometry by using mobile devices in the field. Like mLearn4web, it also consists of three components: an authoring tool, a mobile application and a visualization tool.

The activity itself consists of finding certain pre-defined positions using the GPS sensor of mobile devices. In the preparation phase of the activity, six fixed locations, so-called landmarks, are placed by drag and drop in form of a rectangle on a map. The goal of this activity is to find positions that have certain distances from those landmarks. For example, students are asked to find a point that is 21 meters away from one landmark and 32 meters away from a second landmark. While the students try to find this position, they are able to check the distances with a mobile application.

After the activity, the data with all attempts and results is stored as a set of geolocation data. Thus, in the visualization tool, a map is displayed with markers of all attempts of finding the nearest point. Before the activity, the students were not told that they are doing triangulation exercises, but during our studies, we were able to identify that they were applying concepts connected to triangulation. With the help of our visualization tool the teachers were able to demonstrate this to the students during a reflection session and hence explaining concepts of trigonometry. Thus, the workflow of TriGO is matching the one of the field trip [17]: *preparation*, *activity*, *analysis & reflection.*

C. PELARS

The Practice-based Experiential Learning Analytics Research And Support (PELARS) research project aims to generate, analyze, use, and provide feedback for analytics derived from hands-on project-based learning activities [21].

PELARS aims to generate, analyze, use, and provide feedback for learning analytics by performing hands-on, project-based learning activities. The focus of the PELARS project activities is on learning and making things with physical computing (e.g. Arduino-based projects) that provide learners with opportunities to build and experiment with tangible technologies and digital fabrication.

A core part of the PELARS system is the learners' self-documentation for planning, documenting, and reflecting on their work. We applied parts of the mLearn4web project to provide the personal mobile and web tool for the documentation. The personal mobile tool is an application that allows the learners to send rich media and text as a live stream of data into the PELARS learning analytics system. Furthermore, researchers and observers use it to make notes about the progress of the activity. With the help of the authoring tool of mLearn4web this tool was designed to fit the requirements of the PELARS project. However, it was necessary to adapt the authoring tool as well as the mobile application slightly to the new requirements. In order to do so, we added a new module to the authoring tool: a matrix. This matrix allows designing a questionnaire by applying the Likert scale. Researchers and observers used this module to document the progress of the activities on their mobile devices. Furthermore, an additional layer of activity organization was introduced to the mobile documentation application. Each activity consists of three sub-categories, thus the mobile application of mLearn4web at first was not suitable for PELARS. However, a small adjustment of the mobile application allowed us to easily adapt to the requirements of PELARS.

A visualization tool developed by visualizations experts from the PELARS team presents the generated data. Due to the nature of the data, the visualization tool of mLearn4web was not suitable to present these kinds of data.

D. Implementation details

Matching the requirements to provide cross-platform solutions, the majority of the components in the mentioned projects are realized using web technologies exclusively. An exception is the mobile component of the TriGO project. TriGO is a successor of the GEM [22] project and we decided to use the already existing native application since the dedicated mobile application has a Java-based self-adaptive multi-agent system that only runs on Android phones [23]. The authoring tools and the visualization tools of the projects are mainly supposed to be executed on desktop-like computers, that are available in classrooms, and therefore are implemented using HTML5 and JavaScript technologies, in particular AngularJS. The mobile components of mLearn4web and PELARS are also realized using pure web technologies. In this case, HTML5 together with the Ionic framework was utilized to provide a look and feel of a mobile application and increase the user experience. Since the Ionic framework is based on AngularJS, we are staying consistent with the chosen technologies. As backend technologies, NodeJS and MongoDB are used. Besides the scalable nature of MongoDB, it is also a perfect fit for data persistence, since the communications in this system happens in the form of JSON objects and MongoDB accepts JSON object as input and thus, it is possible to handle JSON objects in all layers of our system (frontend, backend, and database). This eliminates the need for processing or object-relational mapping. The non-blocking event-driven I/O architecture of NodeJS guarantees a scalable foundation for our system. With the mentioned technologies, we apply the MEAN (MongoDB, ExpressJS, AngularJS, NodeJS) stack, which proved to be a scalable and reliable solution for (mobile) web applications.

The works on those three scenarios lead to the current efforts of defining a reference architecture. We recognized the connection and similarities in all three scenarios and the advantage of having a common foundation. Thus, we are focusing on defining a more general solution that will help designing new software solutions in the field of mobile learning.

IV. A REFERENCE ARCHITECTURE FOR MOBILE LEARNING

One goal of our work is to propose a reference architecture to provide a common ground for software in mobile learning. We do not yet aim to cover the whole field but we are concentrating on a special scenario. To answer the RQ1, we derive our main concept from the case of a *field trip* and therefore the reference architecture is based on the main components: *preparation*, *activity*, and *analysis & reflection* that matches the three phases mentioned by Krepel and DuVall [17]. A crucial element in the reference architecture is the interface that is used for communication between the core components. The scenarios presented before showed that having the components separated from each other is very valuable. In some cases, we could re-use components from a project and replace another. This was easily possible because the components can operate independently from each other and communicate through the interface. Fig. 1 provides an overview of the architecture that serves as a foundation for the presented applications in section III.

As also shown in Fig. 1, we also provide some recommendations in our architecture – marked by the dotted elements. For instance, we recommend using web technologies in all of the core components. We are also suggesting a RESTful API that is used as an interface, since it is currently one of the most common ways to provide an interface and is proven to be very robust.

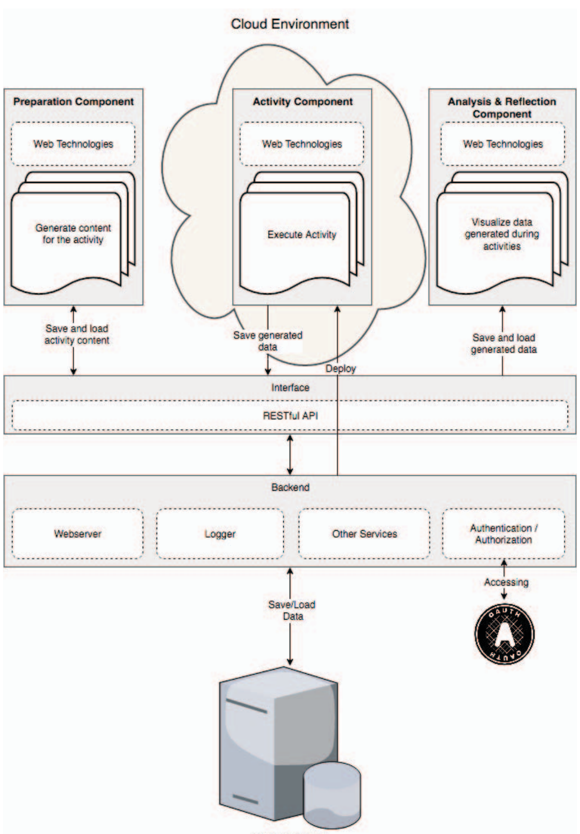

Fig. 1. The architecture used for the mentioned implementations presented in section III

Fig. 1 also emphasizes the role of the interface component. This interface is comparable with an Enterprise Service Bus often used in Service Oriented Architectures (SOA) and thus a crucial element to realize the modularity in our solutions.

The architecture presented in Fig. 1 is derived from the implementations presented in section III. One of the goals of this paper is to define a reference architecture that serves as a foundation for general solutions that deal with similar challenges within the field of mobile learning. To achieve this, it is necessary to generalize this approach; therefore we present in Fig. 2 our reference architecture. The core components (*preparation*, *activity*, and *analysis & reflection*) are independently operating frontend solutions that are mainly communicating via the interface with the backend. Which technologies or approaches in particular are applied for each element depends on the use case/scenario of each individual solution. However, this architecture is designed to support in particular web-based solutions since these satisfy the dynamic requirements in a very effective manner.

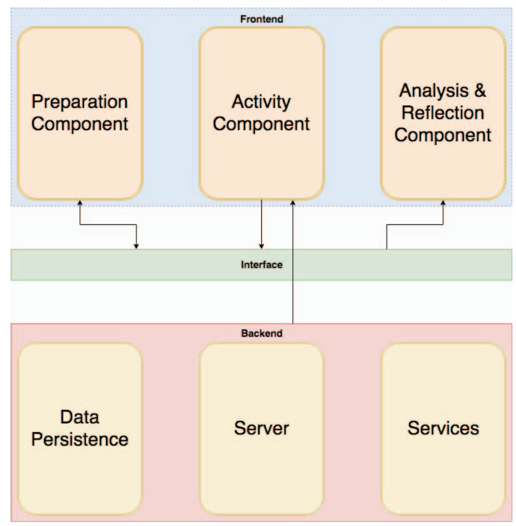

Fig. 2. Reference architecture

V. ARCHITECTURE EVALUATION

One of the purposes of the architecture evaluation of a software system is to analyze and verify that the requirements are met in the design. Thus, to be able to answer RQ2, it is necessary to identify the most relevant system quality attributes that our architecture should possess to satisfy the requirements for the field of mobile learning.

A. *Method of the Architecture Evaluation*

Since there is, to our knowledge, no work yet conducted in evaluating architectures for mobile learning, the first step is to gain an overview on potential methods to perform such an evaluation. The outcomes of this process are at the same time the answers for RQ2.

We chose a scenario-based method [24, 25] for evaluating the architecture since we already have several implementations in place and therefore can describe scenarios in a detailed manner.

The two most relevant options for scenario-based evaluation are: *Software Architecture Analysis Method* (SAAM) [26] and *Architecture Trade-off Analysis Method* (ATAM) [27]. Both are fitting approaches for our needs. For this particular case, we chose to apply the SAAM method because we already defined an architecture and want to evaluate the complete architecture in a qualitative manner. Whereby ATAM is more suited to evaluate tradeoffs between architectural alternatives [25]. One of the capabilities of SAAM is to evaluate the suitability of a given architecture with respect to the desired properties of a specific system. Therefore, this method suits our requirements perfectly.

In the first step of the SAAM method, it is necessary to develop scenarios. We placed our work in the field of mobile learning and also provided three implementation descriptions (see section III). Those implementations, together with the mentioned background serve as scenarios for our evaluation process. In a next step, it is needed to describe the candidate architecture(s). This description can be found in section IV. Thereafter, it is necessary to identify system quality attributes with the help of the developed scenarios. The following sections describe the quality attributes extracted from our scenarios. With the help of these, an overall evaluation of our reference architecture will be conducted.

B. Sytem Quality Attributes

In section II, we discussed that one big challenge for mobile learning activities is the fragmentation of the mobile device market. Therefore, one major quality attribute for an architecture within this field is the *portability*. Software solutions should be available on multiple devices and operating systems to support the bring-your-own-device [5] paradigm.

It was also discussed earlier that the field of mobile learning is evolving in a fast pace. Not only new pedagogical methods are constantly emerging but also the rapid technological enhancements generate very dynamic requirements. New technologies are constantly emerging and applied in educational settings, this ranges from tablets over pervasive displays to multi-touch tables or SMART boards. To cope with those requirements, software solutions should be of an *adaptable* character.

Furthermore, we discussed that mobile learning applications often have similar scenarios, e.g. field trips [17]. Thus, in many cases it is possible to re-use an existing solution. Therefore, *reusability* is important to mention. By mentioning *reusability*, it is also important to acknowledge *extensibility*. Often, only parts of existing solutions are re-used, but to complete a new implementation, it is necessary to either substitute parts of the existing solution or add new elements.

To summarize, the following quality attributes are of special interest for us: *portability*, *adaptability*, *reusability*, and *extensibility*. These software quality attributes match the findings of Vogel [28] who addresses a related challenge. Thus, we are confident that the evaluation of the mentioned attributes is the right direction.

C. Evaluation

SAAM evaluates an architecture by investigating which architectural elements are affected by our scenario. Therefore, in the next step in our evaluation process with SAAM, we evaluate the quality attributes with the help of our implementations. In the following, each of the quality attributes will be analyzed to gain conclusions for the viability of our proposed reference architecture.

1) Portability

In two of our three implementation examples (i.e. mLearn4web and PELARS), all frontend components (*preparation*, *activity*, and *analysis & reflection*) are implemented using web technologies exclusively. As previously discussed, web technologies are a promising approach to address *portability*, also for mobile devices. This shows that the quality attribute of *portability* is satisfied to a high degree with our proposed architecture.

2) Adaptability

The fact that all components can be realized as web applications is also a factor for the quality attribute of adaptability. Due to the fact that web application became widely accepted solutions, emerging technologies have a high probability to support the execution of web applications. In the described implementation of TriGO (see section III.B), we observed that the teachers used a SMART board to visualize the collected data. Furthermore, the mentioned interface is able to provide and consume data from and to any device. This is also demonstrated in the TriGO project by using a native Android application for the activity component instead of a web application. These cases show the *adaptability* of the architecture.

3) Reusability

The three implementation examples serve as examples for the high extent of *reusability* of the proposed architecture. For instance, the preparation component and the activity component of the mLearn4web project and the PELARS project are nearly identical. Most of the elements of mLearn4web are reused in the PELARS project with small adjustments to meet the slightly different requirements. In the PELARS project, a new element was added to the preparation component and a level of interaction was added to the activity component. The new requirements were met with those adjustments showing at the same time the high grade of *reusability* of the architecture.

4) Extensibility

The same example as for the *reusability* can serve as a showcase for the high degree of *extensibility*. The main parts of the preparation and activity components of the mLearn4web project were reused in the PELARS project, but minor modifications were implemented. One element was added that allows to support special content to cope with the requirements for the PELARS project. This exemplifies that the architecture can be extended in a high grade. Another example is the exchange of components itself. For the TriGO project, the activity component was exchanged by a native Android application and within the PELARS project, a new reflection component was developed and added by visualization experts.

This demonstrates the *extensibility* level of the architecture even further.

VI. Conclusion and Outlook

In this paper, we proposed a reference architecture to design software to support mobile learning activities. RQ1 is tackled by introducing the three components: *preparation*, *activity – analysis & reflection* and we present three projects where those are successfully realized. Because of the already existing scenarios, the scenario-based SAAM method was chosen for the evaluation. The outcome of this evaluation strengthens our confidence that this reference architecture is a good foundation for mobile learning software (in field trips scenarios).

During the evaluation process, certain system quality attributes were extracted to perform the evaluation. It is important to mention that this list of attributes is probably not the final list. In the future, different attributes are likely to emerge and further evaluations need to be conducted with new scenarios. Furthermore, we are planning to validate the outcomes in sessions with other experts in the field.

In our coming efforts, we intend to generalize this reference architecture even further. We developed the architecture in the field of mobile learning with the underlying definition of field trips, which resulted in the three core components. We believe that this scenario however is not limited to field trips and the field of mobile learning. By reflecting on the features that have been developed, we believe that we can apply this architecture in other domains. Hence, we plan to apply this reference architecture in other fields/domains that deal with similar problems and challenges. Currently, we are collaborating with a company to develop a system based on this architecture. This also means that additional evaluation will be needed. Due to the change of scenarios it is necessary to update the relevant system quality attributes and again assess those features.

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